

The Chaplygin Sleigh for Every Signed Mass Offset: Exact Scattering, Physical Reconstruction, and the Recurrence Boundary

Route-A structural certificate C199

Abstract

We give a convention-complete theorem for the classical Chaplygin sleigh with mass $m > 0$, central inertia $J > 0$, and arbitrary signed longitudinal offset a . For $a \neq 0$, every non-equilibrium positive-energy orbit is an explicit heteroclinic: its blade rotates through an energy-independent angle and its contact point has two rigorously defined asymptotic lines. We also classify the stable equilibrium half-axis, half-plane Poisson structure, necessarily singular reduced density and its configuration-Haar lift, reversor, and all zero-offset and zero-angular-velocity boundaries. A finite exact/high-precision ledger checks conventions but is not used to prove the continuous theorem. The Poisson form gives only a formal Route-A hint; the absence of an intrinsic prime carrier and target determinant forces rejection.

Keywords: Chaplygin sleigh; nonholonomic mechanics; heteroclinic scattering; reconstruction; invariant measure; reversibility.

中文摘要

本文对质量、转动惯量均为正且质心纵向偏置可取任意符号的经典 Chaplygin sleigh 给出完整定理。偏置非零时，全部非平衡正能轨道均为显式异宿轨道；刀刃转角与能量无关，接触点具有两条严格定义的渐近直线。本文还统一分类稳定平衡半轴、半平面泊松结构、必然奇异的不变测度、时间反演以及零偏置等全部边界。有限精确计算只校验约定，不代替连续参数证明；路线最终仍被拒绝。

关键词：非完整力学；异宿散射；重构；不变测度；时间反演。

1 Convention, equations, and ownership

Let $r = (x, y)$ be the knife contact point, $\theta \in \mathbb{S}^1$ the *blade* angle, u the signed contact velocity along the blade, and $\omega = \dot{\theta}$. Let $a \in \mathbb{R}$ be the signed longitudinal offset from contact point to centre of mass and put $I_c = J + ma^2$. In physical time, the reduced equations and reconstruction are

$$\dot{r} = ue(\theta), \quad \dot{\theta} = \omega, \quad \dot{u} = a\omega^2, \quad \dot{\omega} = -\frac{ma}{I_c}u\omega, \quad e(\theta) = (\cos \theta, \sin \theta). \quad (1)$$

They imply

$$H = \frac{m}{2}u^2 + \frac{I_c}{2}\omega^2 = \text{constant}. \quad (2)$$

The model, its nonholonomic reduction and Hamiltonization context are classical [1, 2, 3]. We claim a signed-parameter synthesis and certificate, not priority for those results.

2 All nonzero-offset scattering orbits

Theorem 1 (signed-offset normal form). *Assume $a \neq 0$, $H > 0$, and $\omega(0) \neq 0$. Define*

$$R = \sqrt{2H/m}, \quad A = \frac{m|a|R}{I_c}, \quad \eta = \frac{\sqrt{I_c/m}}{|a|}, \quad \sigma = \text{sgn } \omega(0).$$

There are unique $t_0 \in \mathbb{R}$ and θ_0 modulo 2π such that, with $s = A(t - t_0)$,

$$u = \text{sgn}(a)R \tanh s, \quad \omega = \sigma R \sqrt{m/I_c} \text{sech } s, \quad \theta = \theta_0 + \sigma \eta \arcsin(\tanh s). \quad (3)$$

It joins $(-\text{sgn}(a)R, 0)$ to $(\text{sgn}(a)R, 0)$ and its blade-angle deflection is

$$\Delta\theta = \theta_+ - \theta_- = \sigma\pi\eta, \quad (4)$$

which is independent of H .

Proof. The sign of a nonzero ω cannot change by uniqueness. On the energy ellipse write $u = \text{sgn}(a)R \tanh s$ and $\omega = \sigma R \sqrt{m/I_c} \text{sech } s$. The first reduced equation gives $\dot{s} = A$; substitution gives the second. Since $d\{\arcsin(\tanh s)\}/ds = \text{sech } s$, integration of $\dot{\theta} = \omega$ proves (3). Taking $s \rightarrow \pm\infty$ proves both endpoints and (4). $\square$

The endpoint table makes the signed convention explicit:

	$t \rightarrow -\infty$	$t \rightarrow +\infty$
u	$-\text{sgn}(a)R$	$\text{sgn}(a)R$
ω	0^σ	0^σ
θ	$\theta_0 - \sigma\eta\pi/2$	$\theta_0 + \sigma\eta\pi/2$

3 Contact-point reconstruction and two lines

Put $q = \tanh s \in (-1, 1)$. Since $dt = dq/[A(1 - q^2)]$, the full $SE(2)$ reconstruction reduces exactly to

$$r(q) - r(0) = \frac{\text{sgn}(a)R}{A} \int_0^q \frac{y \mathbf{e}(\theta_0 + \sigma \eta \arcsin y)}{1 - y^2} dy. \quad (5)$$

This is a one-dimensional quadrature, not a claim of an elementary primitive.

Proposition 1 (affine asymptotes). *Let $v_{\pm} = u_{\pm} \mathbf{e}(\theta_{\pm})$. There are unique vectors $b_{\pm}$ such that*

$$r(t) = b_{\pm} + v_{\pm} t + o(1) \quad (t \rightarrow \pm\infty).$$

Thus the contact trajectory scatters between two oriented affine lines.

Proof. As $s \rightarrow +\infty$, $u - u_+ = O(e^{-2s})$ while $\theta - \theta_+ = O(e^{-s})$; hence $u \mathbf{e}(\theta) - v_+ = O(e^{-s})$, integrable in physical time because $s = At + O(1)$. Therefore $r(t) - v_+ t$ has a finite limit. The negative end is identical. Uniqueness follows by subtracting two candidate intercepts. $\square$

The blade angle is not always the velocity heading. The physical heading is θ when $u > 0$ and $\theta + \pi$ when $u < 0$ (modulo 2π). Therefore (4) must not be silently relabelled as a velocity-heading deflection.

4 Stability, Poisson form, measure, and reversibility

The entire line $\omega = 0$ is reduced equilibrium. At $(u_*, 0)$ its linearization has eigenvalues 0 and

$$\lambda_{\perp} = -\frac{ma}{I_c} u_*. \quad (6)$$

Hence precisely $au_* > 0$ is transversely stable. On either open half-plane,

$$\{u, \omega\} = \frac{a}{I_c} \omega \quad (7)$$

is a nondegenerate Poisson bracket and $\dot{f} = \{f, H\}$ reproduces the reduced flow. The density $\rho = 1/|\omega|$ satisfies $\nabla \cdot (\rho \dot{u}, \rho \dot{\omega}) = 0$ there.

No positive C^1 reduced invariant density can cross a nonzero reduced equilibrium: at $(u_*, 0)$, invariance would give $0 = \rho \nabla \cdot F = -\rho m a u_*/I_c$, a contradiction. Thus the reduced singularity is structural, not a removable normalization. Multiplying by configuration Haar volume gives an invariant full-flow measure off the singular line, and no smooth density of this Haar-factor form crosses it. However, $(u_*, 0)$ with $u_* \neq 0$ still reconstructs as translation in the full five-dimensional flow. The pointwise reduced argument therefore does not exclude every possible configuration-dependent full-flow C^1 density.

Finally,

$$\mathcal{R}(r, \theta, u, \omega) = (r, \theta, -u, -\omega)$$

satisfies $F(\mathcal{R}z) = -D\mathcal{R} F(z)$, so it is a physical-time reversor.

5 All degenerate boundaries and recurrence

If $a \neq 0$ but $\omega_0 = 0$, the reduced state is fixed and r is a straight line; $H = 0$ is the fully fixed case. Otherwise $\dot{u} = a\omega^2$ is strictly one-signed, excluding nonconstant recurrence.

If $a = 0$, both u and ω are constant. When $\omega \neq 0$,

$$r(t) - r(0) = \frac{u}{\omega} (\sin(\theta_0 + \omega t) - \sin \theta_0, -\cos(\theta_0 + \omega t) + \cos \theta_0),$$

and the complete $SE(2)$ state is periodic with period $2\pi/|\omega|$. When $\omega = 0$, motion is straight (fixed if also $u = 0$). Thus $a = 0$ is a sharp recurrence boundary, not a regular member of the scattering family.

Revision focus. Round 1 adds physical reconstruction, affine asymptotes, stability, Poisson/measure obstruction, reversibility and every degenerate recurrence boundary.

Declarations

Data and code availability. Package-local synthetic exact evidence and deterministic code accompany the manuscript; no observational data are used.

Ethics. No human, animal, personal or sensitive data are used; approval is not applicable.

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